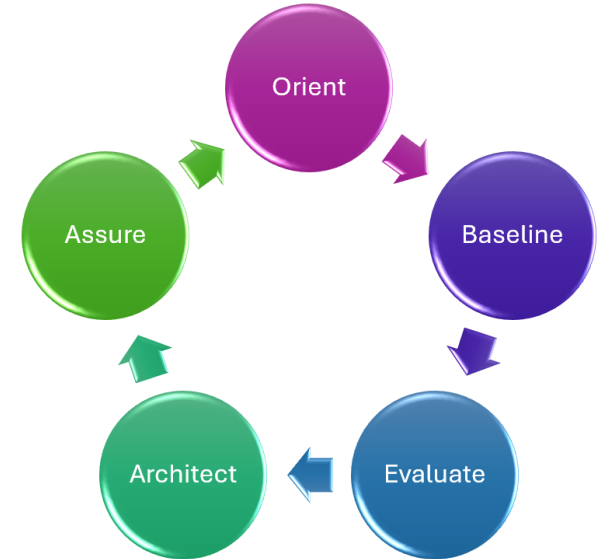


Sustainable Digital Architecture Framework

diagrams



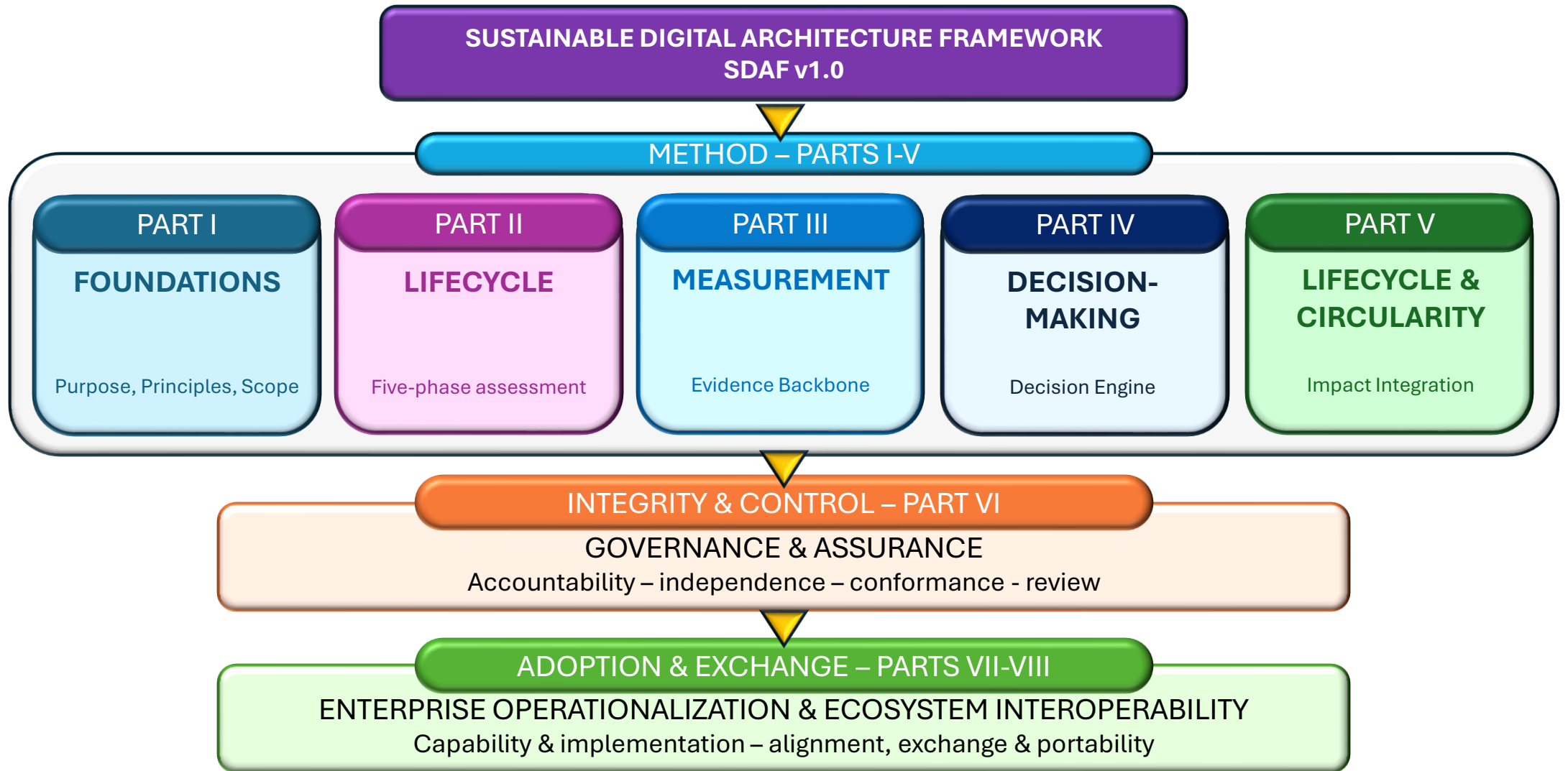
Editable diagram library for SDAF v1.0

Concept provenance

The lifecycle, metrics, and circularity perspectives in this framework build on established sustainability and energy-efficiency practices and have been refined through practical architectural work, lifecycle analysis, and evidence-driven decision-making. The methodology integrates lifecycle assessment (LCA), Scope 2 considerations, embodied-impact perspectives, and audit-ready evidence models to support consistent, standards-aligned architectural decisions. The intent is to provide a reusable foundation that can inform management systems, operations, and delivery contexts while keeping methodology ownership distinct from execution accountability.

SDAF ARCHITECTURE – PARTS I – VIII

One framework: method, integrity, adoption, exchange



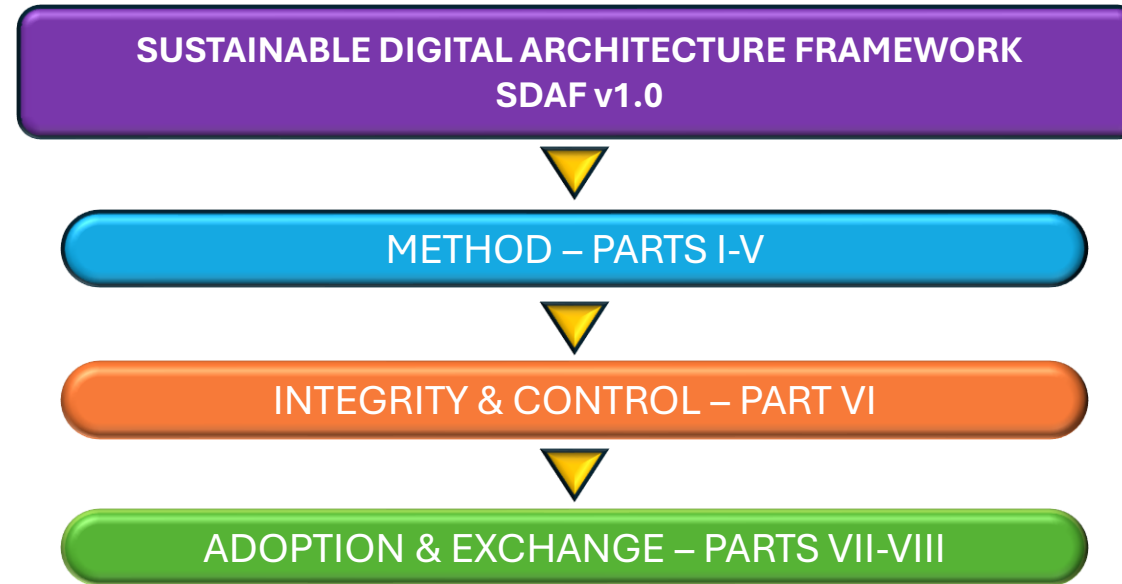
The parts remain distinct; together they form one controlled architecture

Figure 1 — SDAF Architecture (Parts I–VIII)

The overall structure of the Sustainable Digital Architecture Framework, showing how Method, Integrity, and Adoption layers form one controlled architecture.

SDAF ARCHITECTURE – PARTS I – VIII

One framework: method, integrity, adoption, exchange



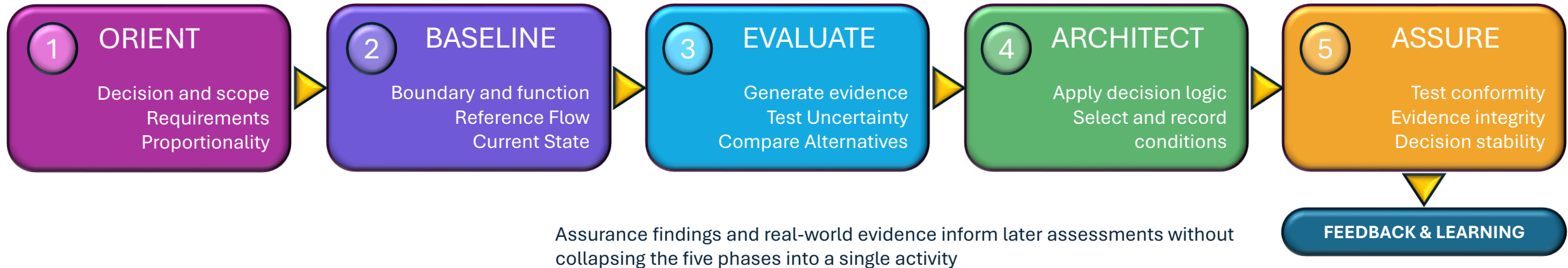
The parts remain distinct; together they form one controlled architecture

Figure 1a — Compact SDAF Architecture

The overall structure of the Sustainable Digital Architecture Framework, showing how Method, Integrity, and Adoption layers form one controlled architecture.

SDAF FIVE-PHASE LIFECYCLE

The repeatable backbone of every SDAF assessment



The lifecycle is sequential, but not static.

Each phase produces controlled outputs for the work that follows. Review triggers may reopen an earlier phase when assumptions, evidence, requirements or operating conditions materially change.

CONTROLLED PROGRESSION

EXPLICIT HAND-OFFS

RECORDED RATIONALE

REVIEW WHEN CONDITIONS CHANGE

Figure 2 — SDAF Five-Phase Lifecycle

The repeatable backbone of every SDAF assessment, illustrating the sequential yet adaptive progression from orientation to assurance and learning.

SDAF Measurement Backbone

How raw inputs become decision-grade evidence

RAW INPUTS



SDAF MEASUREMENT BACKBONE

1 BOUNDARY

What is included?

2 FUNCTION

What service is compared?

3 METHOD

How is it quantified?

4 QUALITY

How representative is it?

5 UNCERTAINTY

What could change?

6 TRACEABILITY

Can another person reproduce it?

DECISION-GRADE EVIDENCE

Comparable – qualified – reproducible – fit for purpose

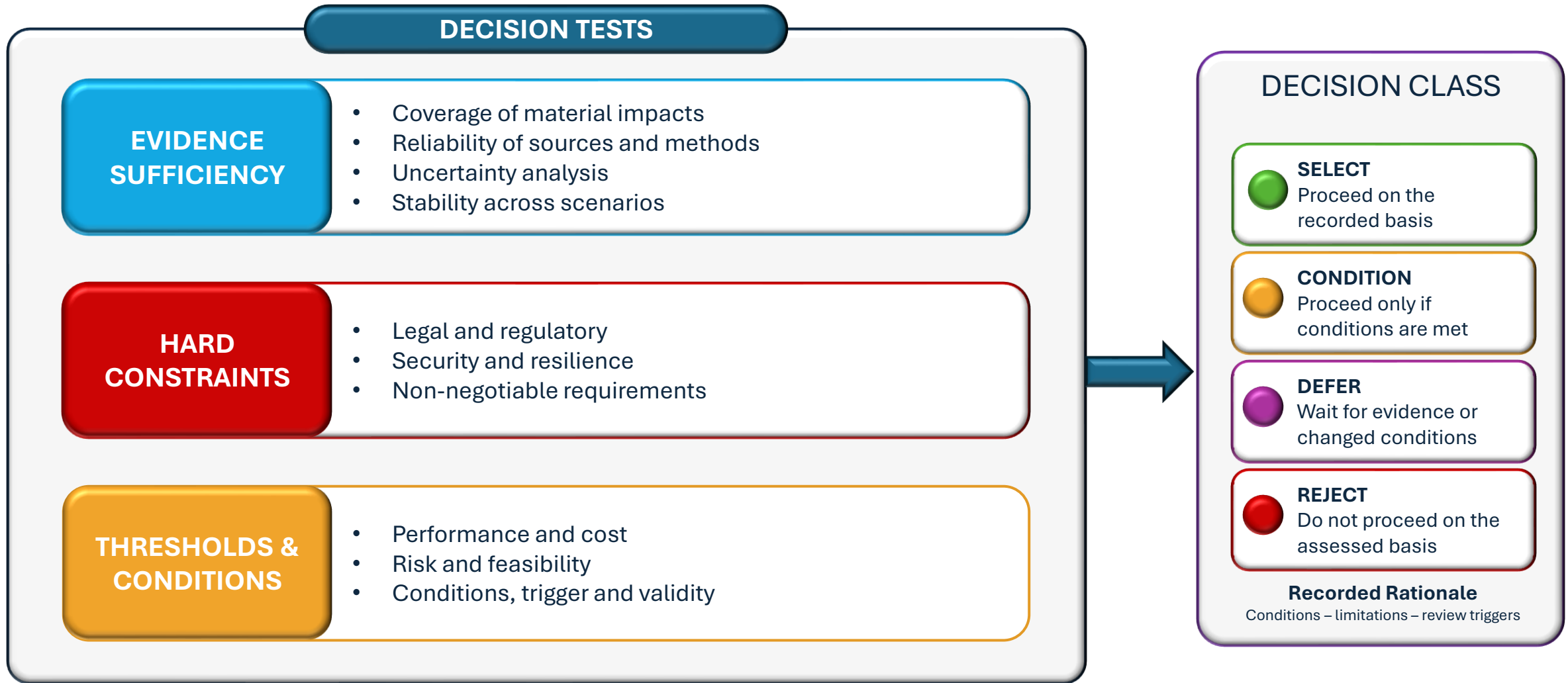
Numbers become evidence only after explicit measurement controls are applied

Figure 3 — SDAF Measurement Backbone

How raw inputs become decision-grade evidence through controlled measurement boundaries, quality, uncertainty, and traceability checks.

SDAF Decision Engine

How evidence becomes a defensible decision



A decision class expresses what the evidence and constraints support – not a sustainability label

Figure 4 — SDAF Decision Engine

The logic that transforms evidence into defensible decisions, showing decision classes and the tests for sufficiency, constraints, and stability.

Lifecycle & Circularity Integration

Burdens and benefits remain separate until controlled decision integration

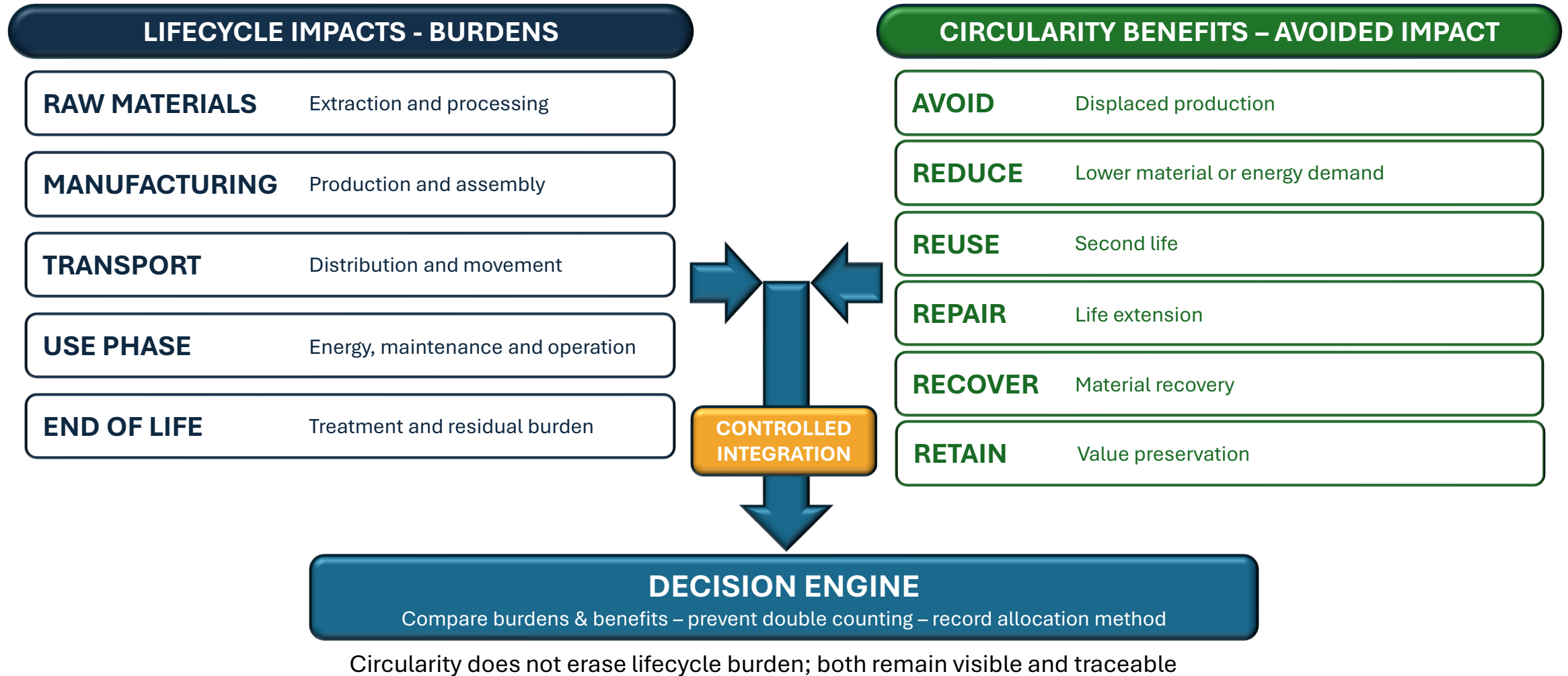


Figure 5 — Lifecycle & Circularity Integration

Controlled integration of lifecycle burdens and circularity benefits, ensuring both remain visible and traceable within the decision process.

SDAF Governance & Assurance

Distinct authorities protect methodological integrity

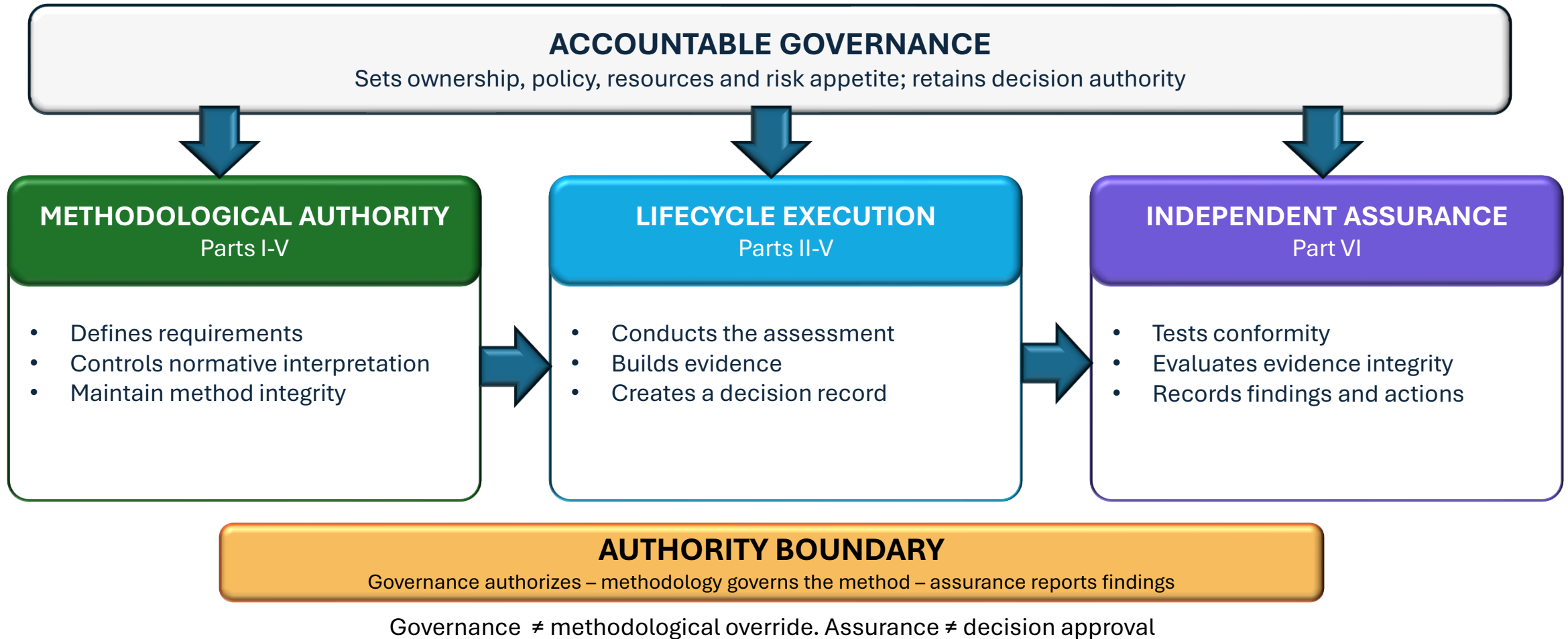
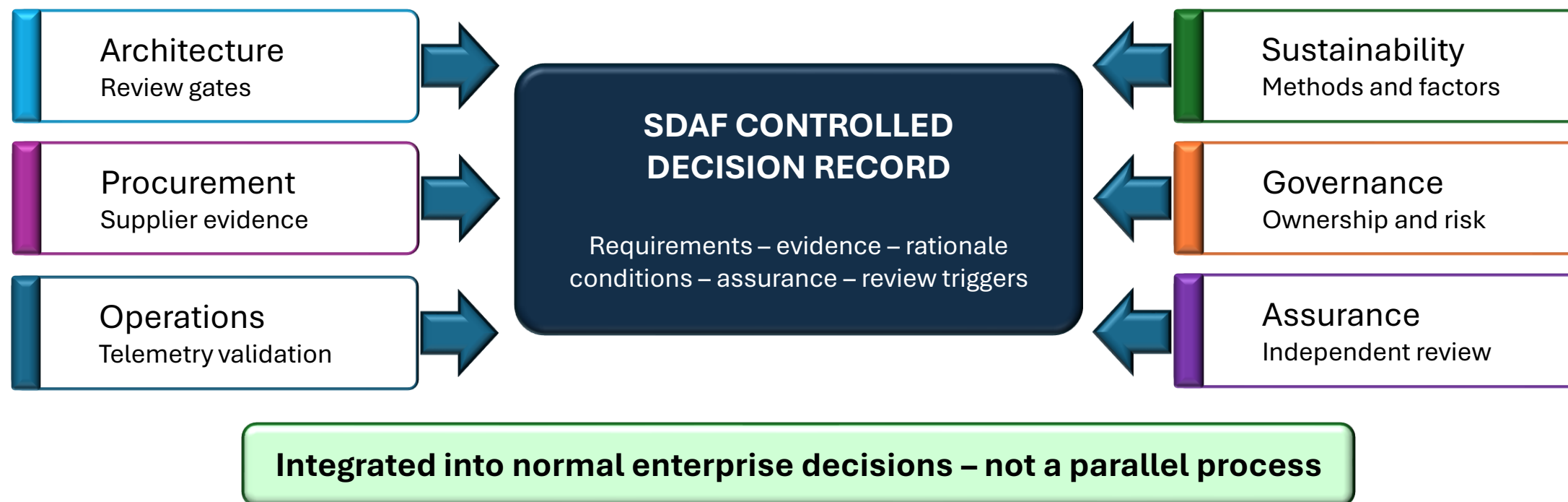


Figure 6 — SDAF Governance & Assurance
Distinct authorities that protect methodological integrity, clarifying the boundaries between governance, lifecycle execution, and independent assurance.

Enterprise Operationalization

SDAF integrates into existing enterprise functions



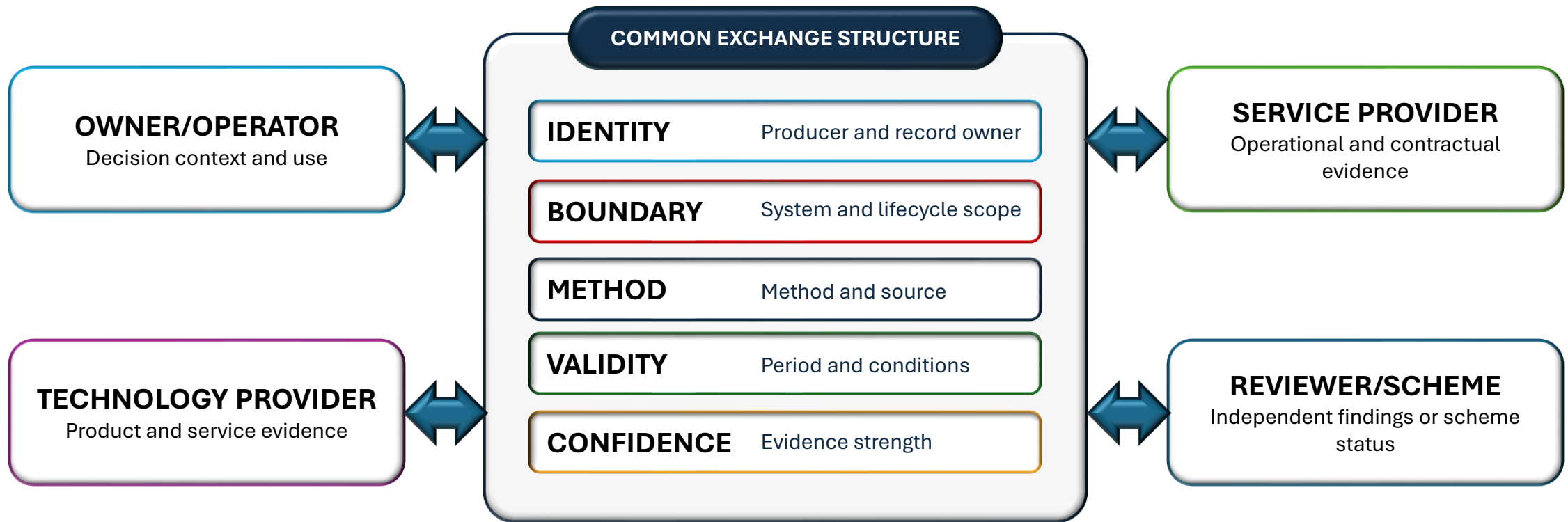
Each function contributes through its existing authority, records and review points

Figure 7 — Enterprise Operationalization

How SDAF integrates into existing enterprise functions, linking architecture, procurement, operations, sustainability, governance, and assurance through a controlled decision record.

Ecosystem Interoperability

Exchange evidence without collapsing provenance or authority



Shared structure enables exchange while provenance and authority remain explicit

Interoperability enables alignment and portability without turning unlike evidence into the same evidence

Figure 8 — Ecosystem Interoperability

The common exchange structure that enables evidence sharing across organizations while preserving provenance, authority, and confidence.